

ANTOINE ASPEEL | CV

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Antoine Aspeel

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»» Professional Summary

I hold a Ph.D. in Mathematical Engineering with a focus on control theory and optimization, and my research centers on safety control and networked control systems. Over the years, I have published work on topics such as safety control, Koopman theory, reachability analysis, networked control systems, and resource-aware control. My aim is to contribute to the development of mathematical and algorithmic tools for designing controllers that guarantee the safety properties of large-scale systems. A key aspect of my research is the design of resource-aware controllers for networked systems, ensuring both performance and safety specifications. During my Ph.D. and postdoc experiences, I worked on multiple team projects and pursued personal research, while also mentoring several master and Ph.D. students. I am excited about the opportunity to bring my expertise to a new team and continue my research in an open-minded setting.

»» Skills

Areas of Expertise

Hybrid systems

Safety control

Optimization

Formal methods

Triggered control

Predictive control

Software / programming

Python ●●●●●●

Julia ●●●●●●

Matlab ●●●●●●

LEAN ●●●●●●

Languages

French ●●●●●●

English ●●●●●●

Italian ●●●●●●

»» Education

2024 - 2025

Postdoc in Control and OptimizationCentraleSupélec, Gif-sur-Yvette,
France

» With **Antoine Girard** and **Thiago Alves Lima** in the *Laboratoire des Signaux et Systèmes* (L2S). Focus on formal methods for safety-critical control.

2022 - 2024

Postdoc in Control and OptimizationUniversity of Michigan, Ann Arbor,
United States of America

» With **Prof. Necmiye Ozay**. Focus on safety control, formal methods, Koopman theory, reachability analysis, and resource-aware control.

2022

Postdoc in Control and OptimizationUniversité Catholique de Louvain,
Louvain-la-Neuve, Belgium

» With **Prof. Raphaël M Jungers** and **Prof. Benoit Macq**. Focus on networked control systems, and resource-aware control.

2017 - 2022

Ph.D. in Mathematical EngineeringUniversité Catholique de Louvain,
Louvain-la-Neuve, Belgium

» Thesis title: "Optimal Sampling for State Estimation of Stochastic Dynamical Systems". Supervisors: **Prof. Raphaël M Jungers** and **Prof. Benoit Macq**.

» From May to June 2019, I was a visiting PhD student at McGill University, Montreal, Canada. I was working with **Prof. Vincent François-Lavet** on deep reinforcement learning.

- » Advanced courses in dynamical systems, control, optimization and graph theory.
- » Thesis title: "Community detection in large-scale time-varying networks: a modularity based approach". Supervisor: **Prof. Jean-Charles Delvenne**.
- » International exchange in 2016 at the École Polytechnique Fédérale de Lausanne (EPFL), Switzerland.

- » I have completed the credits for the double bachelor's degree in applied mathematics and mechanical engineering.

»» Publications

I have published scientific articles in control journals and conferences (check my Google Scholar profile [here](#)).

Articles published in scientific journals

- » Mohamad Louai Shehab, **Antoine Aspeel**, Necmiye Ozay. Learning Reward Machines from Partially Observed Optimal Policies. *Transactions on Machine Learning Research*, 2025.
- » Ram Padmanabhan, **Antoine Aspeel**, Necmiye Ozay, Melkior Ornik. Mode-Prefix-Based Control of Switched Linear Systems with Applications to Fault Tolerance. *IEEE Control Systems Letters*, 2025.
- » Michaël Fanuel, **Antoine Aspeel**, Michaël T. Schaub, Jean-Charles Delvenne. Ellipsoidal embeddings of graphs. *SIAM Journal on Applied Mathematics*, 2024.
- » **Antoine Aspeel**, Jakob Nylof, Jing Shuang (Lisa) Li, and Necmiye Ozay. A Low Rank Approach to Minimize Sensor-to-Actuator Communication in Finite Horizon Output Feedback. *IEEE Control Systems Letters*, 2023. (With presentation at ACC24).
- » Haldun Balim, **Antoine Aspeel**, Zexiang Liu, and Necmiye Ozay. Koopman-inspired Implicit Backward Reachable Sets for Unknown Nonlinear Systems. *IEEE Control Systems Letters*, 2023. (With presentation at CDC23).
- » **Antoine Aspeel**, Amaury Gouverneur, Raphaël M Jungers, and Benoit Macq. Optimal intermittent particle filter. *IEEE Transactions on Signal Processing*, 2022.
- » Michaël Fanuel, **Antoine Aspeel**, Jean-Charles Delvenne, and Johan AK Suykens. Positive semi-definite embedding for dimensionality reduction and out-of-sample extensions. In *SIAM Journal on Mathematics of Data Science*, 2022.
- » Damien Dasnoy, **Antoine Aspeel**, Kevin Souris, and Benoit Macq. Locally tuned deformation fields combination for 2D cine-MRI-based driving of 3D motion models. In *Physica Medica*, 2022.
- » **Antoine Aspeel**, Axel Legay, Raphaël M Jungers, and Benoit Macq. Optimal measurement budget allocation for Kalman prediction over a finite time horizon by genetic algorithms. *EURASIP Journal on Advances in Signal Processing*, 2021.

Works in proceedings of conferences

- » **Antoine Aspeel**, Antoine Girard, Thiago Alves Lima. Exploiting Over-Approximation Errors as Preview Information for Nonlinear Control, 2025. (Under review)
- » **Antoine Aspeel**, Laurent Bako, and Necmiye Ozay. Minimal L2-Consistent Data-Transmission. 2024 63rd IEEE Conference on Decision and Control (CDC), 2024. (With presentation at the **invited session** on "Event-Triggered and Self-Triggered Control").
- » **Antoine Aspeel**, and Necmiye Ozay. A Simulation Preorder for Koopman-like Lifted Control Systems. Presentation in The 8th IFAC Conference on Analysis and Design of Hybrid Systems, 2024.
- » Mohamad Louai Shehab, **Antoine Aspeel**, Nikow Aréchiga, Andrew Best, and Necmiye Ozay. Learning True Objectives: Linear Algebraic Characterizations of Identifiability in Inverse Reinforcement Learning. In the 6th Learning for Dynamics and Control Conference, 2024.
- » **Antoine Aspeel**, Kwesi Rutledge, Raphaël M Jungers, Benoit Macq, and Necmiye Ozay. Optimal control for linear networked control systems with information transmission constraints. In *The 60th IEEE International Conference on Decision and Control*, 2021.

» **Antoine Aspeel**, Amaury Gouverneur, Raphaël M Jungers, and Benoit Macq. Optimal measurement budget allocation for particle filtering. In *27th IEEE International Conference on Image Processing*, 2020.

» **Antoine Aspeel**, Damien Dasnoy, Raphaël M Jungers, and Benoit Macq. Optimal intermittent measurements for tumor tracking in X-ray guided radiotherapy. In *Medical Imaging 2019: Image-Guided Procedures, Robotic Interventions, and Modeling*, volume 10951, page 109510C. International Society for Optics and Photonics, 2019.

»»» Service to scientific community

I was the **co-chair** of the **invited session** of the Conference on Decision and Control (CDC) 2024 on *Event-Triggered and Self-Triggered Control* and I was the **chair** for the session *Control over Communications* at the CDC 2021.

I serve in the **IEEE CSS Technical Committee** on Robust and Complex Systems (TCRoCS).

I am also a **reviewer** for the following journals and conferences (check my web of science profile [here](#)):

- » TAC — IEEE Transactions on Automatic Control.
- » Automatica, Elsevier Journal.
- » EURASIP Journal on Advances in Signal Processing.
- » L-CSS — IEEE Control Systems Letters.
- » EJCON — EUCA European Journal of Control.
- » CDC — IEEE Conference on Decision and Control.
- » ACC — IEEE American Control Conference.
- » ECC — EUCA European Control Conference.
- » L4DC — Conference on Learning for Dynamics and Control.
- » ADHS — IFAC Conference on Analysis and Design of Hybrid Systems.
- » ICCPS — ACM/IEEE International Conference on Cyber-Physical Systems.
- » AAAI — Association for the Advancement of Artificial Intelligence.

»»» Proposal Writing and Mentoring

» I **co-authored a research proposal** on reinforcement learning **that received 1.1 M€** from the Walloon region of Belgium.

» I mentored two visiting Master's students and one Ph.D. student at the University of Michigan, as well as one Master's student from UCLouvain. **Each of these mentoring activities resulted in scientific publications.**

» I serve as a **member of the Ph.D. advisory committee** for a Ph.D. student at UCLouvain.

» Supervised and was a **member of the jury for 6 master's theses** at UCLouvain.

»»» Teaching Activities

From 2014 to 2022, I accumulated **200 hours of teaching experience** at UCLouvain, Belgium (including 20 hours of master-level courses). My responsibilities included leading exercise sessions, as well as designing and grading assignments for the following courses:

- » Introduction to Algebra, undergraduate level.
- » Algebra, undergraduate level.
- » Introduction to Calculus, undergraduate level.
- » Calculus, undergraduate level.
- » Discrete mathematics and probabilities, undergraduate level.
- » Economics, undergraduate level.
- » Sustainable development, master level.

»»» Other Activities

- » Between 2014 and 2016, I was a founding member, treasurer and then elected president of a non-profit organization. I was responsible for the management of reusable glasses for student activities (turnover ~ 30,000€/year).
- » I have been an organizer of youth movements.
- » I run regularly, enjoy playing chess and piano, reading and learning new things.